

**AMENDMENT NO. 1 DECEMBER 1995
TO
IS 12590 : 1988 SPECIFICATION FOR GLYCERINE
FOR COSMETIC INDUSTRY**

(Page 1, clause 0.5) — Substitute the following for the existing clause:

'0.5 To check the presence of adulterants such as diethylene glycol, ethylene glycol and propylene glycol, a test method has been provided as Annex B for guidance.'

(Page 9, Appendix A) — Add the following new appendix at the end:

'APPENDIX B

**THIN LAYER CHROMATOGRAPHIC ANALYSIS FOR
DETECTION OF DIETHYLENE GLYCOL, ETHYLENE GLYCOL
AND PROPYLENE GLYCOL AS ADULTERANTS IN GLYCERINE
(Clause 0.5)**

B-0 GENERAL

The presence of diethylene glycol, ethylene glycol and propylene glycol is determined by thin layer chromatographic analysis.

B-1 PROCEDURE

B-1.1 Reagents

B-1.1.1 Acidic Potassium Dichromate Solution

Dissolve 7 g potassium dichromate in 70 ml water, add 10 ml of sulphuric acid slowly. Cool, mix and make up to 100 ml with water. If the solution turns green, it should be discarded.

B-1.2 Method of Analysis

A typical condition for TLC analysis of glycerine is given below:

Test solution of sample : 2% *m/v* in alcohol glycerine — A

Test solution of standard : 2% *m/v* in alcohol glycerine — B

Test solution of standard : 0.2% *m/v* in alcohol diethylene glycol — C

Test solution of standard : 0.2% *m/v* in alcohol ethylene glycol — D

Test solution of standard : 0.2% *m/v* in alcohol propylene glycol — E

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Spot : 5 µl of Test solution of sample — A ,
Standard substance — B,
Standard substance — C,
Standard substance — D, and
Standard substance — E.

Thin layer plate : Silica gel

Mobile Phase : A mixture of 55 percent chloroform, 35 percent acetone and 10 percent alcohol

Run : 10 cm.

Develop the chromatogram until the solvent front has ascended about 10 cm. After removal of the plate, dry it in a current of air, heat at 105°C for five minutes. Cool and spray with acidic potassium dichromate solution. Allow to dry in air for about 15 minutes.

B-3 REPORT

The blue principal spot in the chromatogram obtained with solution A shall correspond to the spot obtained with solution B and there shall be no other additional spot other than the principal spot obtained with solution A corresponding to the spots obtained with solutions C, D and E.'

(PCD 19)